WEB TECHNOLOGIES

Unit 5: PHP Server-Side Scripting & Databases

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UNIT 5: PHP Server-Side Scripting | Arrays, Functions, Forms | Advance PHP | MySQL Databases | Connecting to Server | DB Commands | Table & Data Operations | phpMyAdmin & Database Bugs

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Important Questions

1 PHP - Server-Side Scripting

Definition:

PHP (Hypertext Preprocessor) is a widely-used, open-source, server-side scripting language that is especially suited for web development. It is embedded directly within HTML to create dynamic, data-driven web pages.

Detailed Explanation:

While HTML, CSS, and JavaScript run on the user's browser (Client-Side), PHP runs entirely on the Web

Server (Server-Side). When a user requests a PHP page (like `index.php`), the web server executes the PHP code, connects to databases if necessary, processes the logic, and generates plain HTML. This plain HTML is what is sent back to the user's browser.

Because the code runs on the server, the user can never see the actual PHP source code—only the HTML output. This makes server-side scripting highly secure, allowing developers to hide complex business logic, database passwords, and file structures from the public.

PHP scripts are executed on the server, and the result is returned to the browser as plain HTML. A PHP file normally contains HTML tags, and some PHP scripting code. PHP code is enclosed within `<?php ... ?>` tags.

Example: Basic PHP Script

```
<!DOCTYPE html>
<html>
<body>

<h1>My first PHP page</h1>

<?php
    // Outputting text to the browser
    echo "Hello World! This text is generated by PHP.";
?>

</body>
</html>
```

Key Characteristics of PHP:

- ✓ **Loosely Typed:** You don't have to declare data types for variables (e.g., `$name = "Rahul";`).
- ✓ **Cross-Platform:** Runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.).
- ✓ **Server Compatibility:** Compatible with almost all servers used today (Apache, IIS).
- ✓ **Database Support:** Supports a wide range of databases, predominantly MySQL.

Hinglish Explanation:

PHP *ek aisi language hai jo Web Server par chalti hai (Client ke browser par nahi). Jab aap kisi PHP website par jaate hain, toh server pehle PHP code ko padhta hai, usko process karta hai, aur ek normal HTML page banakar aapke computer ko bhej deta hai. Isliye aap kabhi kisi website ka PHP code "View Source" karke nahi dekh sakte. Ye data ko secure rakhne aur database se connect karne ke liye best hai.*

2 PHP Arrays and Functions

Definition:

An **Array** in PHP is a special variable that can hold more than one value at a time. A **Function** is a reusable block of statements that can be used repeatedly in a program, executing only when it is explicitly called.

PHP Arrays:

Arrays are essential when you want to store a list of items, such as a list of users or products. PHP supports three types of arrays:

- **Indexed Arrays:** Arrays with a numeric index (starts at 0). E.g., `$cars = array("Volvo", "BMW", "Toyota");`
- **Associative Arrays:** Arrays with named keys instead of numbers. E.g., `$age = array("Peter"=>"35", "Ben"=>"37");`
- **Multidimensional Arrays:** Arrays containing one or more arrays (like a table of rows and columns).

PHP Functions:

PHP has over 1000 built-in functions, but you can also create your own custom **User-Defined Functions**. Functions encapsulate logic to make code modular. They are defined using the `function` keyword. They can accept inputs called *Arguments* and can send back outputs using the `return` statement.

Example: Arrays and Functions

```php
<?php
    // 1. Associative Array Example
    $student_marks = array("Rahul"=>"85", "Amit"=>"92", "Neha"=>"78");
    echo "Rahul got " . $student_marks['Rahul'] . " marks.<br>";

    // 2. User-Defined Function Example
    function calculateTotal($math, $science) {
        $total = $math + $science;
        return $total;
    }

    // Calling the function
    $myTotal = calculateTotal(85, 90);
    echo "Total Marks: " . $myTotal;
?>
```

Advantages:

✓ **Arrays:** Group related data efficiently without creating hundreds of variables.

✓ **Functions:** Promote code reusability (DRY principle) and make debugging easier.

Hinglish Explanation:

Array *ek dabba hai jisme aap ek hi naam ke andar bohot saari values (jaise 10 students ke naam) store kar sakte hain. PHP me 'Associative Array' bohot useful hai jisme number ki jagah naam (key) se data store hota hai (jaise "Rahul"=>"85").* ***Functions*** *ek machine ki tarah hain jisme aap code likh ke rakh dete hain, aur jab zaroorat ho usko call kar lete hain, taaki baar-baar same code na likhna pade.*

3 PHP Forms and Advance PHP

Definition:

PHP Forms handle data submitted by users from HTML forms using Superglobal variables like `$_GET` and `$_POST`. **Advance PHP** involves concepts like Sessions, Cookies, and File Handling to maintain user state and manage server files.

Form Handling:

When an HTML form is submitted, the data is sent to the server. PHP catches this data using Superglobals:

- **$_GET:** Data is sent via the URL. It is visible to everyone (e.g., `page.php?name=Rahul`). Best for search queries, but bad for passwords.
- **$_POST:** Data is sent securely in the HTTP body. It is invisible in the URL and has no size limits. Used for login forms and sensitive data.

Advance PHP Concepts:

1. Cookies: Small files stored on the *user's computer* by the server. Used to remember non-sensitive user preferences (like Dark Mode or language). Created using `setcookie()`.

2. Sessions: A way to store information (in variables) to be used across multiple pages. Unlike cookies, the information is stored on the *server* securely. Used heavily for User Login systems. Started with `session_start()`.

3. File Handling: PHP can create, read, write, and delete files on the server using functions like `fopen()`, `fwrite()`, and `fclose()`.

Example: Handling a POST Form & Session

```
<!-- HTML Form -->
<form method="post" action="login.php">
    Username: <input type="text" name="user">
    <input type="submit">
</form>

// PHP Code (login.php)
<?php
    session_start(); // Start the session

    if ($_SERVER["REQUEST_METHOD"] == "POST") {
        // Collecting value from form
        $username = $_POST['user'];

        // Saving to session variable
        $_SESSION["logged_in_user"] = $username;

        echo "Welcome, " . $username;
    }
?>
```

Hinglish Explanation:

*Jab aap koi form (jaise login page) submit karte hain, toh data PHP ke paas `$_POST` (secure) ya `$_GET` (URL me dikhne wala) ke through aata hai. **Advance PHP** mein **Sessions** aate hain. HTTP internet ki ek bhulne wali aadat hai (stateless). Isliye jab aap login karte hain, PHP ek 'Session' start karta hai jo server par yaad rakhta hai ki "Haan, ye user login hai", taaki aapko har naye page par wapas password na daalna pade.*

4 phpMyAdmin and Database Bugs

📌 Definition:

phpMyAdmin is a free, open-source web-based Graphical User Interface (GUI) tool written in PHP, used to manage MySQL and MariaDB databases easily. **Database Bugs** refer to errors, connection failures, or vulnerabilities (like SQL Injection) that occur while PHP interacts with the database.

📖 phpMyAdmin:

Instead of typing complex SQL queries into a black command-line terminal, web developers use phpMyAdmin. It provides a visual dashboard to create databases, design tables, insert data, export/import backups (.sql files), and manage user permissions using clicks instead of code. It is heavily bundled with local server environments like XAMPP and WAMP.

📖 Common Database Bugs & Troubleshooting:

When connecting PHP to MySQL, several common bugs occur:

- **Connection Errors:** "Access denied for user" happens when the username, password, or host in the `mysqli_connect()` function is incorrect.
- **Syntax Errors:** "You have an error in your SQL syntax" occurs when a query is missing a quote, comma, or spells a command incorrectly.
- **Undefined Table/Column:** Calling a column name that doesn't exist in the database schema.
- **SQL Injection (Vulnerability):** A critical security bug where a hacker inputs malicious SQL code into a web form, which the database accidentally executes. (Prevention: Use Prepared Statements).

🔧 Key Features of phpMyAdmin:

✓ Intuitive web interface.

✓ Support for most MySQL features (managing databases, tables, columns, relations).

✓ Ability to execute raw SQL statements visually.

✓ Import and export data in various formats (CSV, SQL, XML, PDF).

👉 ***Hinglish Explanation:***

***phpMyAdmin** ek website jaisa software hai jo database (MySQL) ko handle karna bahut asaan bana deta hai. Isme command line ki zaroorat nahi hoti, aap mouse se click karke table bana sakte hain aur data*

dekh sakte hain. ***Database bugs*** *tab aate hain jab PHP aur Database ka connection fail ho jaye (wrong password) ya aapne SQL query likhne me koi spelling mistake kar di ho. Sabse khatarnak bug 'SQL Injection' hai jisse hackers database hack kar lete hain.*

5 Connecting to Server & Database Management Commands

Definition:

Before PHP can access, insert, or manipulate database records, it must execute a script establishing a secure **Connection to the Database Server**. Database Management commands involve Data Definition Language (DDL) to Create, Select, and List entire databases and their tables.

Connecting to Server:

In modern PHP, we use the `mysqli` (MySQL Improved) extension or PDO (PHP Data Objects) to connect to a MySQL server. The `mysqli_connect()` function requires four parameters: Server name (usually 'localhost'), Username (default 'root'), Password (default ''), and Database name.

Basic Database Commands with PHP:

We pass standard SQL queries through the `mysqli_query()` PHP function to execute them.

- **Creating a Database:** `CREATE DATABASE dbname;`
- **Selecting a Database:** In PHP, you usually select it during the connection. Alternatively, `mysqli_select_db($conn, 'dbname');` is used.
- **Listing Databases:** `SHOW DATABASES;`
- **Listing Tables:** `SHOW TABLES;`

Example: Connecting and Creating a Database

```php
<?php
    $servername = "localhost";
    $username = "root";
    $password = "";

    // 1. Connection to server
    $conn = mysqli_connect($servername, $username, $password);

    // Check connection
    if (!$conn) {
        die("Connection failed: " . mysqli_connect_error());
    }
    echo "Connected successfully!<br>";

    // 2. Creating Database
    $sql = "CREATE DATABASE CollegeDB";
    if (mysqli_query($conn, $sql)) {
```

```
        echo "Database created successfully";
    } else {
        echo "Error creating database: " . mysqli_error($conn);
    }
?>
```

Hinglish Explanation:

Database se baat karne ke liye PHP ko pehle usse phone lagana padta hai, jise hum ***Connection*** *kehte hain. Iske liye hum* `mysqli_connect()` *use karte hain jisme server ka naam aur password dalna padta hai. Ek baar connect hone ke baad, hum* `mysqli_query()` *function ke andar SQL ki commands (jaise* `CREATE DATABASE`*) likh kar database ko command de sakte hain.*

6 Table Operations (Create, Alter, Drop)

Definition:

Table Operations define the schema (structure) of the database. It involves creating new tables to hold data, modifying existing tables to add or remove columns, and deleting entire databases or tables when they are no longer needed.

Creating a Table:

A database table consists of columns and rows. When creating a table, you must define the column names, their Data Types (like `INT` for numbers, `VARCHAR` for text), and Constraints (like `PRIMARY KEY` to uniquely identify rows, or `AUTO_INCREMENT`).

SQL: `CREATE TABLE tablename (col1 type, col2 type);`

Altering Tables:

If you created a 'Students' table but forgot to add an 'Email' column, you don't need to delete the table. You use the `ALTER TABLE` command to modify the schema without losing existing data.

SQL: `ALTER TABLE tablename ADD columnname datatype;`

Deleting Database and Tables:

To completely destroy a table and all its data, the `DROP TABLE` command is used. To destroy an entire database, `DROP DATABASE` is used. *Warning: This action cannot be undone.*

Example: Creating a Table via PHP

```
<?php
    // Connecting directly to specific database
    $conn = mysqli_connect("localhost", "root", "", "CollegeDB");

    // 1. Creating a Table
    $sql = "CREATE TABLE Students (
        id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
        firstname VARCHAR(30) NOT NULL,
```

```
        lastname VARCHAR(30) NOT NULL,
        email VARCHAR(50)
    )";

    if (mysqli_query($conn, $sql)) {
        echo "Table Students created successfully.";
    }

    // 2. Altering Table (Adding a Phone column)
    $alter_sql = "ALTER TABLE Students ADD phone VARCHAR(15)";
    mysqli_query($conn, $alter_sql);

    // 3. Deleting Table / Database (Commented for safety)
    // mysqli_query($conn, "DROP TABLE Students");
    // mysqli_query($conn, "DROP DATABASE CollegeDB");
?>
```

Hinglish Explanation:

Data store karne ke liye pehle ek structure (Table) banana padta hai. `CREATE TABLE` se table banta hai, aur usme batana padta hai ki text aayega (VARCHAR) ya number (INT). Agar table banne ke baad koi naya column add karna ho (jaise Phone Number), toh hum `ALTER TABLE` use karte hain. Pura table ya database delete karne ke liye `DROP` command ka use hota hai.

7 Data Operations & Queries (Insert, Select, Delete)

Definition:

Data Operations (Queries) involve manipulating the actual records inside the database tables using Data Manipulation Language (DML). It includes adding new data, retrieving specific data, and deleting specific records.

Inserting Data:

To add new rows into a table, the `INSERT INTO` command is used. You specify the table, the columns you are filling, and the values.

Queries (Selecting Data):

The `SELECT` statement is the most used query. It fetches data from the database. You can select all columns using `SELECT *` or specific columns. Using the `WHERE` clause allows you to filter the results (e.g., Select students where city='Delhi'). In PHP, after executing a Select query, we use the `mysqli_fetch_assoc()` function inside a `while` loop to extract and print the rows one by one.

Deleting Data:

The `DELETE FROM` command is used to remove existing records in a table. It is crucial to use a `WHERE` clause when deleting. If you omit the WHERE clause, all records in the table will be permanently deleted!

Example: Insert, Select, and Delete via PHP

```php
<?php
    $conn = mysqli_connect("localhost", "root", "", "CollegeDB");

    // 1. Inserting Data
    $insert_sql = "INSERT INTO Students (firstname, lastname, email)
                   VALUES ('Rahul', 'Sharma', 'rahul@example.com')";
    mysqli_query($conn, $insert_sql);

    // 2. Selecting / Querying Data
    $select_sql = "SELECT id, firstname, lastname FROM Students";
    $result = mysqli_query($conn, $select_sql);

    // Fetching data row by row
    if (mysqli_num_rows($result) > 0) {
        while($row = mysqli_fetch_assoc($result)) {
            echo "ID: " . $row["id"]. " - Name: " . $row["firstname"]. "<br>";
        }
    }

    // 3. Deleting Data
    // Deletes the student whose ID is 1
    $delete_sql = "DELETE FROM Students WHERE id=1";
    mysqli_query($conn, $delete_sql);
?>
```

Key SQL Commands Recap:

- ✓ **INSERT:** `INSERT INTO table (col1, col2) VALUES ('val1', 'val2');`
- ✓ **SELECT:** `SELECT * FROM table WHERE col = 'val';`
- ✓ **DELETE:** `DELETE FROM table WHERE id = 1;`

Hinglish Explanation:

*Table banne ke baad usme data dalne ko **Insert** kehte hain. Data wapas screen par dikhane ke liye **Select (Query)** ka use hota hai. Agar koi record (jaise kisi student ka naam) hatana ho toh **Delete** ka use karte hain. Delete karte waqt hamesha `WHERE` lagana chahiye (jaise jiska Roll Number 5 ho usko delete karo), warna galti se poori table ka saara data delete ho jayega!*

QUICK REVISION NOTES

1. PHP Server-Side:

Executes on the server, generates plain HTML for the browser. Highly secure because source code is hidden. Loosely typed and cross-platform.

2. Arrays, Functions & Forms:

Arrays store multiple values (Indexed, Associative, Multidimensional). Functions promote code reuse. Forms are handled via Superglobals: `$_GET` (URL visible) and `$_POST` (secure/invisible).

3. Advance PHP:
Cookies store small info on the client's browser. **Sessions** store info securely on the server across multiple pages (used for login).

4. DB Connection & Setup:
Use `mysqli_connect()`. Require Host, Username, Password. Commands: `CREATE DATABASE`, `SHOW DATABASES`.

5. Table Operations (Schema):
`CREATE TABLE` (make new), `ALTER TABLE` (add/modify columns), `DROP TABLE` (destroy table completely).

6. Data Queries (Rows):
`INSERT INTO` (add data), `SELECT` (fetch data using fetch_assoc), `DELETE FROM ... WHERE` (remove specific rows).

7. phpMyAdmin & Bugs:
GUI to manage MySQL visually. Common bugs include connection denied, syntax errors in SQL strings, and dangerous SQL Injection attacks.

IMPORTANT QUESTIONS (6 Marks Each)

Q1. What is Server-Side Scripting? Explain the advantages of PHP and demonstrate how PHP processes form data using $_POST and $_GET. 6 Marks

Expected Answer Points:

- Define Server-Side vs Client-Side.
- Advantages of PHP (secure source code, database friendly, cross-platform).
- Explain $_GET (data in URL, less secure, size limit).
- Explain $_POST (data in body, secure, used for passwords). Provide a brief code snippet.

Q2. Differentiate between Sessions and Cookies in PHP. Provide basic syntax for creating and accessing both. 6 Marks

Expected Answer Points:

- Cookies: Stored on client browser, max 4KB, less secure (user prefs). Syntax: `setcookie()`.
- Sessions: Stored securely on the server, ends when browser closes, used for authentication. Syntax: `session_start(); $_SESSION['key']="val";`.

Q3. Write a complete PHP script to connect to a MySQL server, create a database named 'University', and create a table named 'Students' inside it. 6 Marks

Expected Answer Points:

- `$conn = mysqli_connect('localhost', 'root', '');`
- `mysqli_query($conn, "CREATE DATABASE University");`
- Reconnect with DB selected: `mysqli_select_db($conn, 'University');`
- `mysqli_query($conn, "CREATE TABLE Students(id INT, name VARCHAR(50))");`

Q4. Explain Data Definition Language (DDL) operations in PHP. Provide SQL queries and PHP execution methods for Altering and Dropping tables. 6 Marks

Expected Answer Points:

- Define DDL (modifying database structure, not data).
- Altering Table: `ALTER TABLE table_name ADD column_name datatype;`
- Dropping Table: `DROP TABLE table_name;`
- Mention using `mysqli_query($conn, $sql);` to execute them in PHP.

Q5. Write PHP code to insert a record into a table, and then write a SELECT query to fetch and display all records on the webpage. Explain the use of mysqli_fetch_assoc(). 6 Marks

Expected Answer Points:

- Insert Query: `INSERT INTO table (col) VALUES ('val');` executed via `mysqli_query`.
- Select Query: `$res = mysqli_query($conn, "SELECT * FROM table");`
- Explain that `mysqli_fetch_assoc()` fetches the result row as an associative array.
- Show the `while($row = mysqli_fetch_assoc($res)) { echo $row['col']; }` loop structure.

These notes are comprehensive and exam-ready

Practice the PHP connection and query codes as they are highly likely to be asked in exams!

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These notes are created for educational purposes to help students master Web Technologies.